

Two Bugs, Two Fingerprints: Identifying Data-Handling Errors from the Cross-Section of Reported Factor Performance

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Abstract

Look-ahead and survivorship bias are known to inflate backtested performance. The standard treatment of both is prescriptive: it names the error and shows how to avoid it in one's own pipeline. That framing assumes the reader controls the code. The situation that recurs in practice is the opposite. An allocator reads a manager's factor table, a researcher reads a published result, a desk evaluates a vendor's signal. In each case the biased artifact is visible and the pipeline that produced it is not.

We measure what two common data-handling errors do to a twenty-two factor cross-sectional study built on point-in-time US equity data, holding everything else fixed. Joining fundamentals on fiscal period end rather than filing date inflates mean information coefficient by 59% among the eleven factors that read a filed figure, and manufactures four t -statistics above 2.0 from factors that are insignificant under correct dating. Conditioning the universe on current index membership does not uniformly inflate. Mean IC *falls* by 0.0043, while `amihud_illiquidity` flips sign and becomes significant ($t : -1.11 \rightarrow +2.80$) and `turnover_1m` becomes significant with the wrong sign ($t : -0.48 \rightarrow -2.53$).

The two distortions are nearly orthogonal across the factor cross-section (correlation -0.088) and move value factors in opposite directions. Under dating bias, eleven price-and-volume factors are unchanged to machine precision, exactly rather than approximately, because they never read a filed figure. We argue that this structure makes the *pattern* of reported factor performance diagnostic of the underlying error, and we set out the conditions under which that inversion would be valid. Those conditions are not yet tested, and we say so.

1 Introduction

That look-ahead bias inflates backtested returns has been known since Banz and Breen [1]. That survivorship bias does the same is textbook. Neither observation is a contribution, and this paper does not claim otherwise.

What is less established is *how* each error distorts a factor study: which factors move, by how much, and in which direction. This matters because the consumer of factor research is rarely in a position to audit the pipeline. A Sharpe ratio carries no information about which bug produced it. A cross-section of twenty-two factor t -statistics might.

This paper studies two specific data-handling errors as empirical objects. Our central question is:

Given a table of reported factor performance, and no access to the pipeline that produced it, can the underlying data-handling error be identified from the pattern of which factors are distorted?

We answer the forward half of that question and set out conditions for the inverse. Our contributions are:

1. We quantify what each of two errors does to the same factor study, with identical prices, universe, tradability screen and forward returns, changing one thing at a time. Eleven factors that cannot respond to the dating treatment serve as controls and confirm the harness introduces nothing: their IC is identical to machine precision across all three dating conventions.
2. We show the two distortions are structurally distinct. They are nearly orthogonal across the factor cross-section, they disagree in sign on value factors, and one leaves an entire block of factors provably untouched.
3. We characterise the survivorship channel as a forward-looking growth filter rather than a survival filter, and show it relocates rather than inflates measured performance.
4. We propose an inversion—inferring the error from the reported cross-section—and state the four conditions that would have to hold before it could be claimed. Those conditions are not tested here.

The fourth is a proposal rather than a result. We are explicit about that separation throughout, because the first three are measurements and the fourth is not.

2 Data and construction

Universe. S&P 500 point-in-time membership, reconstructed as dated constituent spells from the revision history of a public reference page. This is a free approximation to a point-in-time constituent file and is not CRSP; Section 6 treats the consequences.

Prices. Daily adjusted closes, 2010–2026, with a tradability screen on price, volume and index membership.

Fundamentals. SEC XBRL company facts, keyed on the date each figure was filed. Restatements resolve to the earliest filing. Cumulative cash-flow spans are differenced to true quarters. Rows whose filing date is an XBRL-adoption artifact, meaning a period ending long before the filing that first reports it, are flagged rather than silently trusted.

Factors. Twenty-two published cross-sectional factors, each signed to its published direction, sector-neutralised and rank-normalised. Eleven read a filed fundamental. Eleven use only price and volume.

Evaluation. Information coefficient against 21-day forward returns, out-of-sample from 2014-01-01, with a purged and embargoed walk-forward. The evaluation code is shared across every condition below, so the standard cannot drift between arms.

3 Bug 1: joining fundamentals on fiscal period end

A quarter ending 31 March is not public on 31 March. It becomes public when the 10-Q is filed, a median of 34 days later. Joining on period end hands the backtest a month of information nobody had.

We compare three conventions, holding everything else fixed:

- **PIT** — visible the day the filing landed, and correct.
- **LAG45** — every company files 45 days after quarter end, which is common practice.
- **NAIVE** — visible the instant the quarter closed, which is the error.

3.1 The controls hold exactly

Eleven factors read no filed figure. Across all three conventions their IC is identical, with maximum absolute drift 0.0. That is exact equality in floating point, not a small number.

This is what licenses the rest of the section. It establishes that nothing in the harness responds to the filing calendar except the factors that must.

One factor required care. `turnover_1m` is built from price and volume but divides by shares outstanding, so it inherits the filing calendar through its denominator. It is the only price-family factor that does. We determine dating sensitivity by measurement and check it against a declared set, so a factor that quietly acquires a filed input is surfaced rather than hidden among the controls.

3.2 What the error is worth

Convention	Mean IC inflation	Look better	Cross $t = 2$ spuriously
Fixed 45-day lag	+9% of true IC	8 of 11	0 of 11
Period-end join	+59% of true IC	10 of 11	4 of 11

Table 1: Effect of each dating convention, summarised over the eleven factors that read a filed figure. Averaging in the eleven controls, which cannot move by construction, would halve the reported inflation for free.

The four manufactured significances are `earnings_yield`, `cash_flow_yield`, `roe` and `accruals`. Each is insignificant under correct dating and above $t = 2.0$ under the period-end join.

The comparison between the two rows of Table 1 is the practical result. A fixed 45-day lag is wrong, being right on average and wrong for every filing that arrives later than that, but it manufactures no false discoveries in this sample. The period-end join manufactures four. The distinction between “approximately right” and “wrong” is the distinction between a bias and a bug.

4 Bug 2: conditioning the universe on current membership

Download today’s constituents, pull their price history, backtest. The resulting panel conditions on index membership as of the end of the sample.

4.1 The dominant channel is not the one usually named

Survivorship is normally described as dropping losers: the acquired, the bankrupt, the demoted. In index research that is the smaller half.

Cohort	Names/day	Median \$ volume	Median market cap
Genuine members	386	\$95M	\$10.1B
Wrongly included , not yet admitted	203	\$28M	\$3.5B
Wrongly excluded , later deleted	88	\$54M	\$5.3B

Table 2: Daily averages, 2010–2015. The biased panel wrongly includes more than twice as many names as it wrongly excludes, and the added names are roughly a third the size and a third the liquidity of a genuine constituent.

These are companies that were small early in the sample and grew enough to be admitted later. Conditioning on current membership is therefore not principally a survival filter. It is a forward-looking growth filter aimed at the small and illiquid cohort.

4.2 The distortion relocates rather than inflates

Mean IC change across all twenty-two factors is -0.00432 , and only 8 of 22 improve. Two factors cross $|t| = 2$.

Factor	PIT IC	Biased IC	PIT t	Biased t
amihud_illiquidity	-0.0105	+0.0224	-1.11	+2.80
turnover_1m	-0.0047	-0.0232	-0.48	-2.53
idio_vol_60d	+0.0059	-0.0179	0.49	-1.53
vol_60d	-0.0013	-0.0234	-0.08	-1.49

Table 3: The four factors most affected by conditioning on current membership. Illiquidity flips sign and becomes significant; turnover becomes significant with the wrong sign; the low-volatility anomaly inverts.

In a survivorship-biased panel, being illiquid predicts returns, because the illiquid names in today’s index are precisely those that were small years ago and then grew into it. A biased panel says: buy the illiquid, high-turnover, high-volatility names. That is not a risk premium. It is a description of a company on its way into the index, read backwards.

5 The two fingerprints

The distortions do not overlap. Figure 1 shows both shift vectors on a common axis, and Table 4 summarises them by factor family.

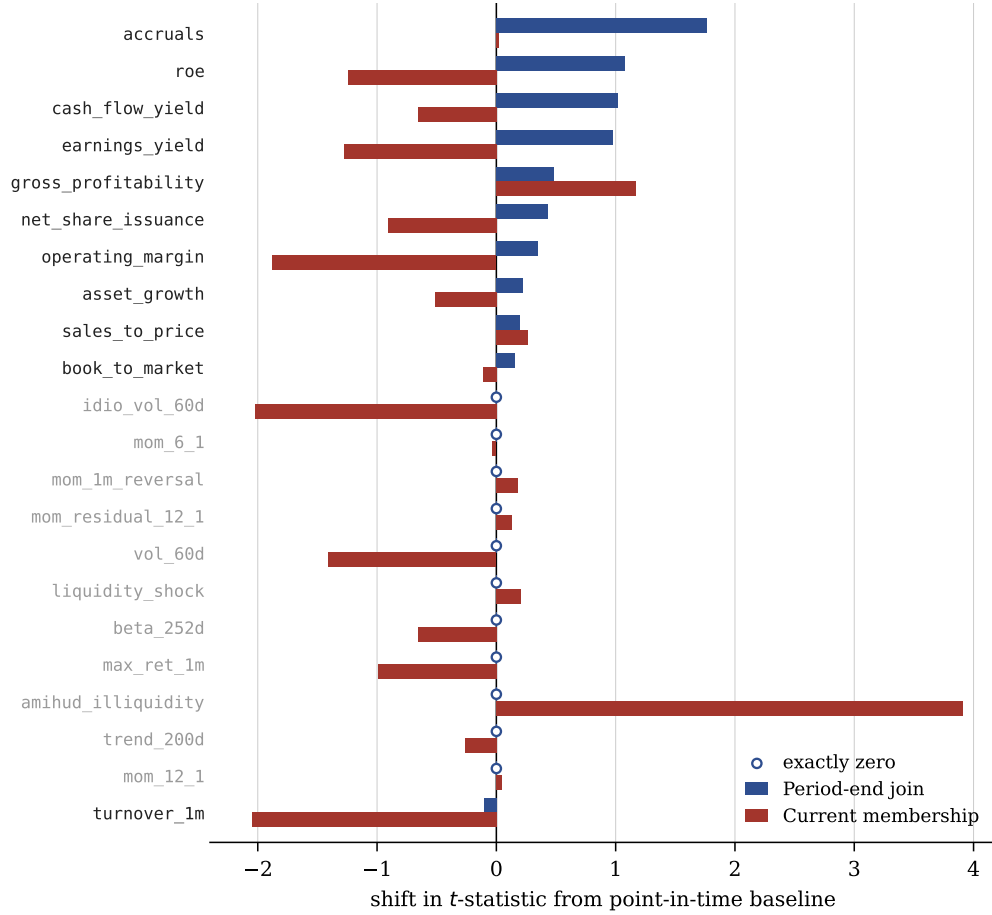


Figure 1: Shift in t -statistic from the point-in-time baseline, all twenty-two factors, ordered by dating shift. Factor names in dark type read a filed fundamental; those in grey use only price and volume. The eleven price-only factors carry an open marker rather than a bar on the period-end series: their shift is exactly zero, to machine precision. The survivorship series has no zeros anywhere, and its largest movement, `amihud_illiquidity` at +3.91, belongs to a factor that dating bias cannot touch.

Quantitatively, the two shift vectors across the twenty-two factors correlate at -0.088 . They are not two versions of “results look better”; they are close to independent directions in factor space. They also disagree in sign on value: dating inflates `earnings_yield` and `roe` by $+0.98$ and $+1.07$ t -units, survivorship deflates them by -1.28 and -1.24 .

One property is a hard constraint rather than a tendency. Under dating bias the eleven price-only factors are unchanged to machine precision, because they never read a filed figure. An anomalously strong illiquidity result therefore cannot be produced by dating bias, whatever else is true of the study.

5.1 The proposed inversion, and what it would require

This structure suggests reading the implication backwards. A study with a suspiciously strong illiquidity premium and a dead low-volatility anomaly has a universe problem. A study whose value and quality factors clear $t = 2$ while its price factors look ordinary has a dating problem.

Factor family	Period-end join	Current membership
Price and volume	Untouched, exactly	Heavily distorted
Value and quality	Inflated; 4 false positives	Mixed, mostly weakened
Liquidity and volatility	Untouched	Sign flips, false positives
Direction of effect	Uniform inflation	Relocation, not inflation

Table 4: The two distortions by factor family.

Formally, model an observed table of t -statistics as

$$t = \tau + \delta_b + \varepsilon, \quad (1)$$

where τ is the study’s true factor performance, δ_b is the signature of bias b (measured, with $\delta_{\text{none}} = 0$), and ε is sampling noise. Since τ is unknown and twenty-two dimensional, b is not identified without further structure. Three sources of structure are available, in increasing order of assumption: exclusion from the exact-zero constraint of Section 3; within-family contrasts that absorb the unknown level of τ ; and projection onto each signature after removing family means, whose reference distribution under exchangeable τ yields a p -value rather than a label.

We state plainly that this inversion is proposed and not tested. Four conditions would have to hold, and each is a gate rather than a caveat:

1. **Signature stability.** The shift vectors must be properties of the errors, not of the 2014–2026 sample. This requires re-measurement on disjoint subperiods and random half-universes.
2. **Signature uncertainty.** A point estimate cannot support a test. The sampling covariance of each shift vector must be estimated.
3. **Separability under realistic noise.** Whether the two errors remain separable at the noise level of a twelve-year study is an empirical question.
4. **Calibration before application.** Any output must be a p -value or a posterior with a reference distribution behind it, never a bare label.

Labelled data for the third gate is free: biased panels are generated by the pipeline itself, so the inversion can be validated at scale without any external dataset or annotation.

6 Limitations

- One index, one country, 2010–2026. The fingerprints are measured on the S&P 500 and should not be assumed to transfer to small caps or non-US markets.
- Membership spells are reconstructed from a public wiki revision history, not CRSP. The survivorship arm inherits whatever that approximation gets wrong.
- Fundamental coverage begins in 2010 because of the XBRL mandate phase-in, so the dating arm cannot speak to the pre-XBRL era.
- The two arms are not additively decomposable. The survivorship arm rebuilds forward returns per universe, because a biased study never observes the returns of the names it excludes. Real studies contain several errors at once and the signatures may not compose.

- A study reporting five factors gives a five-dimensional observation, and the exclusion constraint does not apply at all if none of them are price-only.
- The inversion in Section 5.1 is untested. Sections 3 and 4 are measurements.

7 Conclusion

Two common data-handling errors distort a factor study in different and identifiable ways. The period-end join inflates fundamental factors uniformly by 59% of true IC and manufactures four false discoveries while leaving price factors provably untouched. Current-membership conditioning relocates rather than inflates, flipping illiquidity to significance and inverting the low-volatility anomaly, through a channel better described as a forward-looking growth filter than as a survival filter.

Neither distortion is visible in a Sharpe ratio. Both are visible in the cross-section of what a study reports. Whether that visibility supports reliable inference from a table to its underlying error is the open question, and Section 5.1 states what would have to be established before it could be claimed.

Reproduction. All results reproduce with `make bias`. The dating arm is `scripts/pit_vs_naive.py`; the survivorship arm is `scripts/survivorship.py`. Per-factor outputs are written to `reports/pit_vs_naive.csv` and `reports/survivorship.csv`. Prices are fetched per user rather than redistributed. The point-in-time fundamentals layer and the membership spells are released under CC BY 4.0.

References

- [1] R. Banz and W. Breen. Sample-dependent results using accounting and market data: some evidence. *Journal of Finance*, 41(4):779–793, 1986.